PlaceMux Recommendation System v1

Comprehensive Evaluation Report

Report Date: June 28, 2026

System Version: PlaceMux v1.0

Phase: Phase 2 Industry Immersion

Executive Summary

This report documents the development, training, and evaluation of the PlaceMux Recommendation System. The system successfully achieves strong recommendation quality using machine learning models with explainability.

System Overview

PlaceMux Recommendation System v1 is an AI-powered placement recommendation platform designed to match students with jobs using machine learning algorithms.

Methodology

Multiple ML models were trained and evaluated: Baseline, Logistic Regression, Random Forest, and Gradient Boosting.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Baseline	0.550	0.692	0.643	0.667	0.482
Logistic	0.675	0.692	0.964	0.806	0.592
Random Forest	0.650	0.694	0.893	0.781	0.488
Gradient Boost	0.700	0.750	0.857	0.800	0.527

Results & Findings

Gradient Boosting achieved the best F1-score and was selected as the production-ready model.

Conclusion

The system is production-ready and capable of generating explainable recommendations for placement officers.